Physics 710-711-712 November 16, 2009

Problem Set 5

Problem 1: Do exercise 4.2.1 of the text.

(1)

Solution: The possible outcomes are $L_z = \{1, 0, -1\}$, which are the eigenvalues of L_z.

(2)

Solution: $L_z|\psi\rangle = 1 \cdot |\psi\rangle$ implies

$$|\psi\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

(Note that I have normalized $|\psi\rangle$!) Then

$$\begin{aligned}
\langle L_x\rangle &= \langle\psi|L_x|\psi\rangle = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 0. \\
\langle L_x^2\rangle &= \langle\psi|L_x^2|\psi\rangle = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} \frac{1}{2} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{2}. \\
\Delta L_x &= \sqrt{\langle L_x^2\rangle - \langle L_x\rangle^2} = \sqrt{\left(\frac{1}{2}\right) - 0^2} = \frac{1}{\sqrt{2}}.
\end{aligned}$$

(3)

Solution: The characteristic equation for L_x is

$$0 = \det(L_x - \lambda) = \det \begin{pmatrix} -\lambda & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & -\lambda & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & -\lambda \end{pmatrix} = \lambda - \lambda^3, \qquad \Rightarrow \qquad \lambda \in \{1, 0, -1\}.$$

The corresponding eigenvectors, $|\lambda\rangle$, then satisfy

$$0 = (L_x - \lambda)|\lambda\rangle = \begin{pmatrix} -\lambda & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & -\lambda & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & -\lambda \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} -\lambda a + \frac{b}{\sqrt{2}} \\ \frac{a}{\sqrt{2}} - \lambda b + \frac{c}{\sqrt{2}} \\ \frac{b}{\sqrt{2}} - \lambda a \end{pmatrix}$$

where we have parameterized the components of $|\lambda\rangle$ by $(a \ b \ c)$. For $\lambda = 1$, we can solve for b and c in terms of a, giving $b = \sqrt{2}a$ and $c = a$. We then determine a by normalizing $|\lambda = 1\rangle$:

$$|\lambda = 1\rangle = \begin{pmatrix} a \\ \sqrt{2}a \\ a \end{pmatrix}, \qquad \Rightarrow \qquad 1 = \langle\lambda = 1|\lambda = 1\rangle = \begin{pmatrix} a^* & \sqrt{2}a^* & a^* \end{pmatrix} \begin{pmatrix} a \\ \sqrt{2}a \\ a \end{pmatrix} = 4|a|^2, \qquad \Rightarrow \qquad a = \frac{1}{2}$$

(where I have chosen the arbitrary phase to be 1). Thus, and doing the same thing for $\lambda = 0$ and $\lambda = -1$, gives

$$|\lambda=1\rangle = \frac{1}{2}\begin{pmatrix}1\\ \sqrt{2}\\ 1\end{pmatrix}, \qquad |\lambda=0\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ 0\\ -1\end{pmatrix}, \qquad |\lambda=-1\rangle = \frac{1}{2}\begin{pmatrix}1\\ -\sqrt{2}\\ 1\end{pmatrix}.$$

(4)

Solution: The possible outcomes are $L_x = \{1, 0, -1\}$, which are the eigenvalues of L_x. $|\psi\rangle$ is the normalized eigenstate of L_z with eigenvalue $L_z = -1$, which is

$$|\psi\rangle = \begin{pmatrix}0\\ 0\\ 1\end{pmatrix}.$$

So (here $\mathcal{P}$ stands for "probability of"):

$$\begin{aligned}
\mathcal{P}(L_x = 1) &= |\langle\lambda=1|\psi\rangle|^2 = \left|\frac{1}{2}\begin{pmatrix}1 & \sqrt{2} & 1\end{pmatrix}\begin{pmatrix}0\\ 0\\ 1\end{pmatrix}\right|^2 = \frac{1}{4},\\
\mathcal{P}(L_x = 0) &= |\langle\lambda=0|\psi\rangle|^2 = \left|\frac{1}{2}\begin{pmatrix}1 & 0 & -1\end{pmatrix}\begin{pmatrix}0\\ 0\\ 1\end{pmatrix}\right|^2 = \frac{1}{2},\\
\mathcal{P}(L_x = -1) &= |\langle\lambda=-1|\psi\rangle|^2 = \left|\frac{1}{2}\begin{pmatrix}1 & -\sqrt{2} & 1\end{pmatrix}\begin{pmatrix}0\\ 0\\ 1\end{pmatrix}\right|^2 = \frac{1}{4}.
\end{aligned}$$

(5)

Solution:

$$L_z^2 = \begin{pmatrix}1 & & \\ & 0 & \\ & & 1\end{pmatrix}, \qquad \Rightarrow \qquad \text{the possible outcomes are } L_z^2 = \{0, 1\}.$$

An eigenbasis of the $L_z^2 = 1$ eigenspace is $\{|a\rangle, |b\rangle\}$ with

$$|a\rangle = \begin{pmatrix}1\\ 0\\ 0\end{pmatrix}, \qquad |b\rangle = \begin{pmatrix}0\\ 0\\ 1\end{pmatrix}.$$

Therefore, upon measuring $L_z^2 = 1$, the state collapses to

$$|\psi\rangle \longrightarrow |\psi'\rangle = \frac{(|a\rangle\langle a| + |b\rangle\langle b|)|\psi\rangle}{|(|a\rangle\langle a| + |b\rangle\langle b|)|\psi\rangle|}.$$

But

$$[|a\rangle\langle a| + |b\rangle\langle b|]\,|\psi\rangle = \left[\begin{pmatrix}1\\ 0\\ 0\end{pmatrix}\begin{pmatrix}1 & 0 & 0\end{pmatrix} + \begin{pmatrix}0\\ 0\\ 1\end{pmatrix}\begin{pmatrix}0 & 0 & 1\end{pmatrix}\right]\frac{1}{2}\begin{pmatrix}1\\ 1\\ \sqrt{2}\end{pmatrix} = \begin{pmatrix}1\\ 0\\ 0\end{pmatrix}\frac{1}{2} + \begin{pmatrix}0\\ 0\\ 1\end{pmatrix}\frac{1}{\sqrt{2}} = \frac{1}{2}\begin{pmatrix}1\\ 0\\ \sqrt{2}\end{pmatrix},$$

has norm

$$\sqrt{\frac{1}{2}\begin{pmatrix}1 & 0 & \sqrt{2}\end{pmatrix}\frac{1}{2}\begin{pmatrix}1\\0\\\sqrt{2}\end{pmatrix}} = \frac{\sqrt{3}}{2},$$

so

$$|\psi'\rangle = \frac{1}{\sqrt{3}/2}\frac{1}{2}\begin{pmatrix}1\\0\\\sqrt{2}\end{pmatrix} = \frac{1}{\sqrt{3}}\begin{pmatrix}1\\0\\\sqrt{2}\end{pmatrix}.$$

The probability of $L_z^2 = +1$ is

$$\begin{aligned}\mathcal{P}(L_z^2 = 1) &= \langle\psi|\left(|a\rangle\langle a| + |b\rangle\langle b|\right)|\psi\rangle = |\langle a|\psi\rangle|^2 + |\langle b|\psi\rangle|^2\\ &= \left|\frac{1}{2}\begin{pmatrix}1 & 0 & 0\end{pmatrix}\begin{pmatrix}1\\1\\\sqrt{2}\end{pmatrix}\right|^2 + \left|\frac{1}{2}\begin{pmatrix}0 & 0 & 1\end{pmatrix}\begin{pmatrix}1\\1\\\sqrt{2}\end{pmatrix}\right|^2 = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}.\end{aligned}$$

If we measured L_z the posible outcomes are the eigenvalues L_z, $\{0, \pm 1\}$, with probabilities

$$\begin{aligned}\mathcal{P}(L_z = 1) &= \left|\begin{pmatrix}1 & 0 & 0\end{pmatrix}|\psi'\rangle\right|^2 = \left|\frac{1}{\sqrt{3}}\begin{pmatrix}1 & 0 & 0\end{pmatrix}\begin{pmatrix}1\\0\\\sqrt{2}\end{pmatrix}\right|^2 = \frac{1}{3}.\\ \mathcal{P}(L_z = 0) &= \left|\begin{pmatrix}0 & 1 & 0\end{pmatrix}|\psi'\rangle\right|^2 = \left|\frac{1}{\sqrt{3}}\begin{pmatrix}0 & 1 & 0\end{pmatrix}\begin{pmatrix}1\\0\\\sqrt{2}\end{pmatrix}\right|^2 = 0.\\ \mathcal{P}(L_z = -1) &= \left|\begin{pmatrix}0 & 0 & 1\end{pmatrix}|\psi'\rangle\right|^2 = \left|\frac{1}{\sqrt{3}}\begin{pmatrix}0 & 0 & 1\end{pmatrix}\begin{pmatrix}1\\0\\\sqrt{2}\end{pmatrix}\right|^2 = \frac{2}{3}.\end{aligned}$$

(6)

Solution: In the L_z eigenbasis

$$|L_z = 1\rangle = \begin{pmatrix}1\\0\\0\end{pmatrix}, \qquad |L_z = 0\rangle = \begin{pmatrix}0\\1\\0\end{pmatrix}, \qquad |L_z = -1\rangle = \begin{pmatrix}0\\0\\1\end{pmatrix},$$

write the unknown state as

$$|\psi\rangle = \begin{pmatrix}a\\b\\c\end{pmatrix}.$$

Then

$$\begin{aligned}\mathcal{P}(L_z = 1) = \frac{1}{4} &= |\langle L_z = 1|\psi\rangle|^2 = \left|\begin{pmatrix}1 & 0 & 0\end{pmatrix}\begin{pmatrix}a\\b\\c\end{pmatrix}\right|^2 = |a|^2,\\ \mathcal{P}(L_z = 1) = \frac{1}{2} &= |\langle L_z = 0|\psi\rangle|^2 = \left|\begin{pmatrix}0 & 1 & 0\end{pmatrix}\begin{pmatrix}a\\b\\c\end{pmatrix}\right|^2 = |b|^2,\\ \mathcal{P}(L_z = 1) = \frac{1}{4} &= |\langle L_z = -1|\psi\rangle|^2 = \left|\begin{pmatrix}0 & 0 & 1\end{pmatrix}\begin{pmatrix}a\\b\\c\end{pmatrix}\right|^2 = |c|^2.\end{aligned}$$

The most general solution to these three equations is then

$$a = \frac{1}{2}e^{i\delta_1}, \qquad b = \frac{1}{\sqrt{2}}e^{i\delta_2}, \qquad c = \frac{1}{2}e^{i\delta_3},$$

for some arbitrary phases δ_i, which gives the desired answer.

The δ_i phase factors are *not* irrelevant. For example

$$\begin{aligned}
\mathcal{P}(L_x = 0) &= |\langle \lambda = 0|\psi\rangle|^2 = \left| \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & -1 \end{pmatrix} \frac{1}{2} \begin{pmatrix} e^{i\delta_1} \\ \sqrt{2}e^{i\delta_2} \\ e^{i\delta_3} \end{pmatrix} \right|^2 = \left| \frac{e^{i\delta_1}}{2\sqrt{2}} - \frac{e^{i\delta_3}}{2\sqrt{2}} \right|^2 \\
&= \frac{1}{8}\left(e^{i\delta_1} - e^{i\delta_3}\right)\left(e^{-i\delta_1} - e^{-i\delta_3}\right) = \frac{1}{8}\left(1 - e^{i(\delta_3-\delta_1)} - e^{-i(\delta_3-\delta_1)} + 1\right) \\
&= \frac{1}{4}\left(1 - \cos(\delta_3 - \delta_1)\right),
\end{aligned}$$

so something measurable (a probability) depends on the difference of the phases.

Problem 2: Do exercise 4.2.2 of the text.

Solution:

$$\begin{aligned}
\langle P\rangle &= \langle\psi|P|\psi\rangle = \int_{-\infty}^{\infty} dx \langle\psi|x\rangle\langle x|P|\psi\rangle \\
&= \int_{-\infty}^{\infty} dx\, \psi^*(x)\left(-i\hbar\frac{d}{dx}\right)\psi(x) = -i\hbar\int_{-\infty}^{\infty} dx\, \psi(x)\frac{d\psi(x)}{dx} \\
&= -\frac{i\hbar}{2}\int_{-\infty}^{\infty} dx\, \frac{d}{dx}\left(\psi(x)^2\right) = -\frac{i\hbar}{2}\, \psi^2\big|_{-\infty}^{\infty} = 0
\end{aligned}$$

if $\psi \to 0$ as $|x| \to \infty$.

Alternatively, use the k-basis:

$$\langle P\rangle = \langle\psi|P|\psi\rangle = \int_{-\infty}^{\infty} dk \langle\psi|k\rangle\langle k|P|\psi\rangle = \int_{-\infty}^{\infty} dk\, \hbar k\, \langle\psi|k\rangle\langle k|\psi\rangle = \int_{-\infty}^{\infty} dk\, \hbar k\, \psi^*(k)\psi(k).$$

But

$$\psi(k) = \langle k|\psi\rangle = \int_{-\infty}^{\infty} dx\, \langle k|x\rangle\langle x|\psi\rangle = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} dx\, e^{ikx}\psi(x),$$

therefore

$$\psi(-k) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} dx\, e^{-ikx}\psi(x) = \psi^*(k)$$

since $\psi(x)$ is real. So

$$\langle P\rangle = \int_{-\infty}^{\infty} dk\, \hbar k\psi^*(k)\psi(k) = \int_{-\infty}^{\infty} dk\, \hbar k\psi(-k)\psi(k).$$

and under the change of variables $k \to -k$, this becomes

$$\langle P\rangle = \int_{-\infty}^{\infty} dk\, \hbar(-k)\psi(k)\psi(-k) = -\langle P\rangle,$$

and so $\langle P\rangle = 0$.

Problem 3: Do exercise 2.4.3 of the text.

Solution:

$$\begin{aligned}
\left\langle e^{ip_0x/\hbar}\psi \middle| P \middle| e^{ip_0x/\hbar}\psi \right\rangle &= \int_{-\infty}^{\infty} dx \, \langle e^{ip_0x/\hbar}\psi|x\rangle\langle x|P\middle|e^{ip_0x/\hbar}\psi\right\rangle = \int_{-\infty}^{\infty} dx \, \left(e^{ip_0x/\hbar}\psi(x)\right)^* (-i\hbar)\frac{d}{dx}\left(e^{ip_0x/\hbar}\psi(x)\right) \\
&= -i\hbar\int_{-\infty}^{\infty} dx \, \psi^*(x)e^{-ip_0x/\hbar}\left[\frac{ip_0}{\hbar}e^{ip_0x/\hbar}\psi(x) + e^{ip_0x/\hbar}\frac{d\psi}{dx}\right] \\
&= \int_{-\infty}^{\infty} dx \, \psi^*(x)\, p_0\, \psi(x) - i\hbar\int_{-\infty}^{\infty} dx \, \psi^*(x)\frac{d\psi}{dx} \\
&= p_0\left[\int_{-\infty}^{\infty} dx \, \langle\psi|x\rangle\langle x|\psi\rangle\right] + \left[\int_{-\infty}^{\infty} dx \, \langle\psi|x\rangle\langle x|P|\psi\rangle\right] = p_0\langle\psi|\psi\rangle + \langle\psi|P|\psi\rangle = p_0 + \langle P\rangle.
\end{aligned}$$